DSC5001: Statistical Machine Learning I

Week 10: Classification II: CART

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Decision trees

DSC5001 ▪ Semester A, 2026/27

Conditional Rules

Logistic regression has a linear boundary in the original features.
Machine failure may depend on **vibration**, or on **temperature when vibration is low**.

$$\boxed{\text{Vibration} \rightarrow \text{Temperature} \rightarrow \text{Failure decision}}$$

Main idea

A decision tree expresses conditional rules without manually creating interaction terms.

Binary Splits

At a node, choose one feature x_j and one threshold s:

$$\boxed{x_j \leq s \quad \text{or} \quad x_j > s.}$$

- j: index of the selected feature;
- s: threshold learned from training data;
- each branch sends observations to a child node.

A terminal node is a **leaf** that stores a prediction.

Machine Failure Data

Input features: temperature ($^\circ C$) and vibration (sensor units).
Response: $Y = 1$ for failure, $Y = 0$ for normal.

ID	Temp.	Vib.	Y	ID	Temp.	Vib.	Y
1	45	1.0	0	7	68	2.4	0
2	50	1.2	0	8	70	4.2	1
3	55	2.0	0	9	72	1.6	0
4	58	3.0	1	10	75	3.8	1
5	60	1.4	0	11	80	4.0	1
6	63	3.5	1	12	85	1.8	1

12 machines: six normal and six failures. We will use these data to calculate the splits.

Conditional Interactions

A tree can combine conditions even when neither feature alone is sufficient.

High temperature	High vibration	Failure
No	No	0
No	Yes	1
Yes	No	1
Yes	Yes	0

This is an **XOR pattern**: no single straight line separates the four labeled corners.
Splitting first on temperature and then on vibration can separate all four cases.

The Root Node

The root contains all 12 machines:

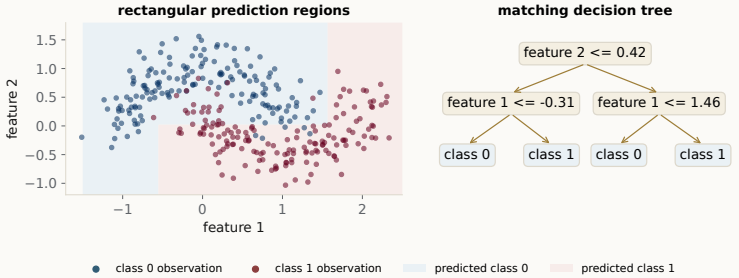
$$n_0 = 6, \qquad n_1 = 6, \qquad n = 12.$$

n_k counts training examples with class k.
The estimated class-one probability is

$$\boxed{\widehat{p}_1 = \frac{n_1}{n} = \frac{6}{12} = 0.50.}$$

Before using features, every machine receives the same estimated probability.
Splitting should create groups with more informative class proportions.

Tree and Regions



Internal node
threshold question

Branch
answer to question

Leaf
stored prediction

Takeaway. A root-to-leaf path is an if–then rule for one prediction region.

Leaf Predictions

Suppose a leaf contains 6 normal machines and 1 failure.

$$\widehat{p}_1 = \frac{1}{7} \approx 0.143, \qquad \widehat{p}_0 = \frac{6}{7}.$$

For equal error costs, predict its majority class:

$$\boxed{\widehat{Y} = 0.}$$

- Classification leaf: class proportions and a class decision.
- Regression leaf: a numerical prediction (later in this lecture).

Leaf Decisions

A leaf with six normal machines and one failure has $\widehat{p}_1 = 1/7$.
Equal error costs: predict normal because $1/7 < 0.5$.
Now suppose false negatives cost 10 and false positives cost 1.
The positive-decision threshold becomes

$$t = \frac{1}{10+1} = \frac{1}{11}.$$

Hence

$$\boxed{1/7 > 1/11 \quad \Rightarrow \quad \text{predict failure.}}$$

The estimated probability is unchanged. Only the **decision rule** changes.

Rectangular Regions

In a two-feature problem:

- a temperature split creates a vertical boundary;
- a vibration split creates a horizontal boundary.

$$\boxed{x_j \le s \quad \Longrightarrow \quad \text{one axis-aligned partition.}}$$

Multiple splits can produce rectangular regions and a nonlinear overall decision boundary.

Main idea

The order of rules matters: a later rule applies only to observations reaching its node.

Candidate Splits

Our root has 6 normal and 6 failed machines.
Consider two possible splits:

Rule	**Left (normal, failure)**	**Right (normal, failure)**
Temp. ≤ 56.5	$(3, 0)$	$(3, 6)$
Vib. ≤ 2.7	$(6, 1)$	$(0, 5)$

We need a numerical measure of **class mixing** to decide which split is better.

Split selection

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Class Mixing

Node	Normal	Failure	Interpretation
A	10	0	pure
B	8	2	mostly normal
C	5	5	strongly mixed

A useful split reduces average class mixing.
Two common measures are

Gini impurity and Entropy.

Node Counts

Let P denote a parent node with n_P observations.

$$\boxed{\widehat{p}_{k,P} = \frac{n_{k,P}}{n_P},}$$

where $n_{k,P}$ counts examples of class k in node P.
A split creates left child L and right child R, with

$$n_P = n_L + n_R.$$

Use class proportions *within each node*.

Gini Impurity

For K classes, the Gini impurity of node P is

$$\boxed{G(P) = 1 - \sum_{k=1}^{K} \widehat{p}_{k,P}^{2}.}$$

For binary classification, let $p = \widehat{p}_{1,P}$:

$$G(P) = 1 - p^2 - (1-p)^2 = \boxed{2p(1-p)}.$$

p	0	0.2	0.5
$G(P)$	0	0.32	0.50

A pure node has Gini 0; a balanced binary node has Gini 0.5.

Gini Interpretation

Draw two class labels independently using a node's empirical class probabilities.
The probability that both labels agree is

$$\sum_{k=1}^{K} \widehat{p}_{k,P}^{2}.$$

Therefore,

$$\boxed{G(P) = \Pr(\text{two independent draws disagree}).}$$

For 80% normal and 20% failure:

$$G(P) = 2(0.8)(0.2) = \boxed{0.32}.$$

Gini therefore measures class disagreement.

Temperature Split

Use the twelve-machine dataset and split at

$$\boxed{\text{Temp.} \le 56.5.}$$

	Normal	Failure	Total
Left	3	0	3
Right	3	6	9

The parent Gini is $G(P) = 0.5$.
The left child is pure:

$$G(L) = 0.$$

The right child has failure proportion $6/9 = 2/3$:

$$G(R) = 2(1/3)(2/3) = \boxed{4/9}.$$

Temperature Gini

A large child should contribute more than a small child.

$$G_{\text{split}} = \frac{n_L}{n_P} G(L) + \frac{n_R}{n_P} G(R).$$

For temperature ≤ 56.5:

$$G_{\text{split}} = \frac{3}{12}(0) + \frac{9}{12}\left(\frac{4}{9}\right) = \boxed{\frac{1}{3}}.$$

The impurity reduction is

$$\boxed{\Delta G_A = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \approx 0.167.}$$

Vibration Split

Now use

$$\boxed{\text{Vib.} \leq 2.7}$$

on the same twelve machines.

	Normal	Failure	Total
Left	6	1	7
Right	0	5	5

The child impurities are

$$G(L) = 1 - (6/7)^2 - (1/7)^2 = \boxed{\frac{12}{49}},$$

$$G(R) = 0.$$

Vibration Gini

With parent size 12:

$$G_{\text{split}} = \frac{7}{12}\left(\frac{12}{49}\right) + \frac{5}{12}(0) = \boxed{\frac{1}{7} \approx 0.143}.$$

Therefore

$$\boxed{\Delta G_B = \frac{1}{2} - \frac{1}{7} = \frac{5}{14} \approx 0.357.}$$

Compare the two candidates:

$$\boxed{\Delta G_B \approx 0.357 > 0.167 \approx \Delta G_A.}$$

The vibration split is preferred.

Gini versus Errors

Gini and training classification error are different criteria.
A separate example has parent class counts $(8, 4)$:

$$G(P) = \frac{4}{9}, \qquad \text{training errors} = 4.$$

Split it into children $(5, 1)$ and $(3, 3)$:

$$G_{\text{split}} = \frac{6}{12}\frac{10}{36} + \frac{6}{12}\frac{1}{2} = \boxed{\frac{7}{18} < \frac{4}{9}}.$$

Yet the minimum number of errors after splitting is

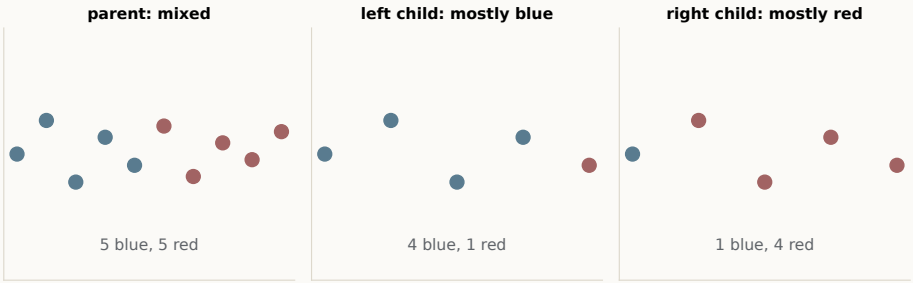
$$\boxed{1 + 3 = 4},$$

unchanged from the parent.

Main idea

Gini can improve even when training misclassification error remains unchanged.

Visualizing Splits



Parent
initial mixing

Children
post-split mixing

Criterion
impurity reduction

Takeaway. A useful split need not make both children pure.

Alternative Vibration Cut

For comparison, try

$$\boxed{\text{Vibration} \le 1.9.}$$

	Normal	Failure	Size
Left	4	1	5
Right	2	5	7

$$G_L = 2(4/5)(1/5) = \frac{8}{25}, \qquad G_R = 2(2/7)(5/7) = \frac{20}{49}.$$

$$\boxed{G_{\text{split}} = \frac{5}{12}\frac{8}{25} + \frac{7}{12}\frac{20}{49} = \frac{13}{35} \approx 0.3714.}$$

This is worse than $1/7$ at the selected cut 2.7.

Candidate Thresholds

For a numerical feature, CART tests thresholds between consecutive distinct sorted values. For vibration, the values surrounding the selected cut are 2.4 and 3.0:

$$\boxed{s = (2.4 + 3.0)/2 = 2.7.}$$

Checking all candidate thresholds gives:

Feature	Best threshold(s)	Minimum child Gini
Temperature	56.5 or 73.5	$1/3$
Vibration	2.7	$1/7$

Thus vibration at 2.7 is the best root split among the available candidates.

Child-Size Weighting

Suppose a split creates:

$$n_L = 90, \qquad n_R = 10.$$

A simple average,

$$\frac{G(L) + G(R)}{2},$$

would give both children equal influence.
Instead CART uses

$$\boxed{G_{\text{split}} = \frac{90}{100} G(L) + \frac{10}{100} G(R).}$$

Larger children contribute more to the total impurity.

Entropy

Motivation: Gini measures label disagreement. Another choice measures uncertainty in the class distribution.

$$\boxed{H(P) = -\sum_{k=1}^{K} \widehat{p}_{k,P} \log \widehat{p}_{k,P}.}$$

Use natural logarithms and $0 \log 0 = 0$.
For binary classification:

$$H(1, 0) = 0,$$

$$H(0.5, 0.5) = \log 2 \approx 0.693.$$

Lower entropy means a more predictable class label.

Temperature Entropy

For temperature ≤ 56.5, the children are $(3, 0)$ and $(3, 6)$.
The first child is pure:

$$H_L = 0.$$

The second has proportions $1/3$ and $2/3$:

$$H_R = -\frac{1}{3}\log\frac{1}{3} - \frac{2}{3}\log\frac{2}{3} \approx \boxed{0.6365}.$$

$$H_{\text{split}} = \frac{3}{12}(0) + \frac{9}{12}(0.6365) \approx 0.4774.$$

$$\boxed{\Delta H \approx 0.6931 - 0.4774 = 0.2158.}$$

Entropy Calculation

For the root with six examples of each class:

$$H(P) = \log 2 \approx 0.6931.$$

For vibration ≤ 2.7, the left child has proportions $6/7$ and $1/7$:

$$H(L) = -\frac{6}{7}\log\frac{6}{7} - \frac{1}{7}\log\frac{1}{7} \approx \boxed{0.4101}.$$

The right child is pure, so $H(R) = 0$.

$$H_{\text{split}} = \frac{7}{12}(0.4101) \approx 0.2392.$$

$$\boxed{\Delta H \approx 0.6931 - 0.2392 = 0.4539.}$$

Gini versus Entropy

For the two root candidates:

Split	Gini decrease	Entropy decrease
Temp. ≤ 56.5	0.1667	0.2158
Vib. ≤ 2.7	0.3571	0.4539

Both criteria select vibration.

Common mistake

Node entropy measures mixing of class labels. It is not the logistic cross-entropy loss from Week 8.

Split Practice

A parent has four normal and four failed observations:

$$G(P) = 0.5.$$

Two candidate splits have class counts (n_0, n_1):

	Left	**Right**
Candidate A	$(3, 1)$	$(1, 3)$
Candidate B	$(4, 2)$	$(0, 2)$

Compute each weighted child Gini and select the better split.

Split Practice Solution

Candidate A
Both children have size 4 and Gini $3/8$:

$$G_A = \frac{4}{8}\frac{3}{8} + \frac{4}{8}\frac{3}{8} = \boxed{0.375}.$$

Candidate B
The left child has Gini $4/9$, and the right child is pure:

$$G_B = \frac{6}{8}\frac{4}{9} + \frac{2}{8}(0) = \boxed{\frac{1}{3} \approx 0.333}.$$

Thus,

$$\Delta G_A = 0.125, \qquad \Delta G_B = \frac{1}{6} \approx 0.167.$$

Choose Candidate B.

Recursive tree growth

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Recursive Splitting

At every nonterminal node:

1. list valid feature-threshold candidates;
2. compute weighted child impurities;
3. choose the largest impurity reduction;
4. repeat within the children.

Root split $\rightarrow$ smaller groups $\rightarrow$ more local rules.

A pure node has Gini 0 and cannot improve its purity further.

Two Root Children

Our selected root rule is

$$\boxed{\text{Vibration} \leq 2.7.}$$

	Normal	Failure	Total
Low vibration	6	1	7
High vibration	0	5	5

The high-vibration child is pure.
The low-vibration child still has Gini

$$\boxed{G = \frac{12}{49}.}$$

Second Split

Only the low-vibration child needs another split.
The normal machines have temperatures

$$45,\ 50,\ 55,\ 60,\ 68,\ 72.$$

The failed machine has temperature 85.
Choose the midpoint:

$$\boxed{s = (72 + 85)/2 = 78.5.}$$

The rule

$$\boxed{\text{Temperature} \leq 78.5}$$

creates a child with six normal machines and a child with one failure.

Equivalent Cuts

Many thresholds can create the same training partition.
At the root, adjacent vibration values are 2.4 and 3.0:

$$\boxed{2.4 \leq s < 3.0}.$$

All these thresholds produce the same split of the observed data.
At the second node, adjacent temperatures are 72 and 85:

$$\boxed{72 \leq s < 85}.$$

The midpoint 78.5 is convenient, not uniquely determined.

Second-Split Gain

Before splitting the low-vibration node:

$$G(P) = 1 - (6/7)^2 - (1/7)^2 = \frac{12}{49}.$$

After splitting at temperature 78.5:

$$G(L) = G(R) = 0.$$

The local impurity reduction is

$$\boxed{\Delta G = \frac{12}{49} \approx 0.245.}$$

Relative to the original twelve-machine dataset:

$$\boxed{\frac{7}{12}\frac{12}{49} = \frac{1}{7} \approx 0.143.}$$

Three-Leaf Tree

Conditions	**Counts** (n_0, n_1)	**Prediction**
Vib. > 2.7	$(0, 5)$	failure
Vib. ≤ 2.7, Temp. ≤ 78.5	$(6, 0)$	normal
Vib. ≤ 2.7, Temp. > 78.5	$(0, 1)$	failure

All twelve training observations are correctly classified.
However, one leaf contains only a single machine.

New Predictions

Apply the rules to three new machines:

Temp.	Vib.	Path	Prediction
60	1.8	low vib., low temp.	normal
82	2.2	low vib., high temp.	failure
65	3.4	high vib.	failure

Each feature-threshold question determines the next branch.
No distance calculation is required.

Greedy Search

At each node, CART selects the best *immediate* split.
At our root, the best split is vibration ≤ 2.7.
Then CART considers new splits separately within the resulting children.

$$\boxed{\text{Locally best rule } \not\Rightarrow \text{ globally best tree.}}$$

Greedy construction is efficient, but does not exhaustively search all possible tree structures.

Greedy Search Limits

Consider four observations with XOR labels:

$$(0,0) \mapsto 0, \quad (0,1) \mapsto 1,$$

$$(1,0) \mapsto 1, \quad (1,1) \mapsto 0.$$

Either root feature split creates two equally mixed children:

$$G_{\text{parent}} = 0.5, \qquad G_{\text{children}} = 0.5.$$

Thus,

$$\boxed{\Delta G = 0.}$$

Yet splitting both children once more produces pure leaves.
A rule requiring a strictly positive immediate gain would stop too early.

Stopping Rules

A tree may stop because of:

- maximum depth;
- minimum leaf size;
- minimum observations needed to split;
- minimum impurity improvement.

In our example, the second split creates leaves of sizes

$$6 \quad \text{and} \quad 1.$$

A minimum leaf size of 2 would prevent this split.

Minimum Leaf Size

With the second split disallowed, the low-vibration leaf retains

$$n_0 = 6, \qquad n_1 = 1.$$

Its prediction becomes

$$\widehat{p}_1 = \frac{1}{7}, \qquad \boxed{\widehat{Y} = 0}.$$

Training errors increase from 0 to 1.

Main idea

One training mistake may be preferable to a rule supported by only one observation.

Impurity Importance

For the full three-leaf tree, global weighted Gini reductions are

$$\text{Vibration: } \frac{5}{14} \approx 0.357,$$

$$\text{Temperature: } \frac{1}{7} \approx 0.143.$$

They sum to the root Gini 0.5.
Normalize to obtain

$$\boxed{\text{Vibration} \approx 71.4\%, \qquad \text{Temperature} \approx 28.6\%.}$$

These are training impurity importances, not causal effects.

Regression trees

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Numerical Responses

CART also applies to continuous outcomes such as energy consumption or house prices. The partitioning mechanism is unchanged:

$$x_j \leq s \qquad \text{versus} \qquad x_j > s.$$

The main differences:

- a leaf predicts a **number**, not a class;
- splits reduce **within-leaf squared error**, not Gini.

Leaf Means

In a leaf R, choose a constant a to minimize

$$Q(a) = \sum_{i \in R} (y_i - a)^2.$$

Differentiate:

$$\frac{dQ}{da} = -2 \sum_{i \in R} (y_i - a) = 0.$$

For n_R observations, the solution is

$$\boxed{\widehat{a} = \frac{1}{n_R} \sum_{i \in R} y_i.}$$

A regression leaf predicts its sample mean.

Regression Parent RSS

Suppose x is a machine setting and y is energy consumption.

x	1	2	3	4	5	6
y	10	12	14	24	26	28

Without splitting, the prediction is

$$\bar{y} = \frac{10 + 12 + 14 + 24 + 26 + 28}{6} = 19.$$

The residual sum of squares is

$$\begin{aligned} RSS_P &= (-9)^2 + (-7)^2 + (-5)^2 + 5^2 + 7^2 + 9^2 \\ &= \boxed{310}. \end{aligned}$$

Regression Split RSS

Split at

$$\boxed{x \leq 3.5.}$$

The responses become

$$R_L : \ (10, 12, 14), \qquad R_R : \ (24, 26, 28).$$

The leaf means are

$$\bar{y}_L = 12, \qquad \bar{y}_R = 26.$$

The within-leaf errors are

$$RSS_L = 4 + 0 + 4 = 8,$$

$$RSS_R = 4 + 0 + 4 = 8.$$

Thus

Alternative Regression Cut

Try the alternative threshold $x \leq 2.5$.
Left responses:

$$(10, 12), \qquad \bar{y}_L = 11, \qquad RSS_L = 2.$$

Right responses:

$$(14, 24, 26, 28), \qquad \bar{y}_R = 23.$$

$$\begin{aligned} RSS_R &= (14-23)^2 + (24-23)^2 \\ &\quad + (26-23)^2 + (28-23)^2 \\ &= 116. \end{aligned}$$

Therefore,

$$\boxed{RSS_{\text{split}} = 2 + 116 = 118.}$$

This is worse than the RSS of 16 at $x = 3.5$.

Regression Cut Comparison

For the same six observations:

Threshold s	2.5	3.5	4.5
Left mean	11	12	15
Right mean	23	26	27
Total RSS	118	**16**	118

Among these candidates,

$$\boxed{s = 3.5}$$

produces the lowest squared error.

Main idea

A useful regression split creates groups with more consistent numerical responses.

Regression Predictions

The selected two-leaf regression model is

$$\boxed{\widehat{f}(x) = \begin{cases} 12, & x \le 3.5, \\ 26, & x > 3.5. \end{cases}}$$

For example,

$$\widehat{f}(2.8) = 12, \qquad \widehat{f}(5.2) = 26.$$

A regression tree is **piecewise constant**.
Further splits create more steps, but can also fit training noise.

Tree complexity and pruning

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Training and Generalization

Our full machine tree has zero training errors.
But one failure leaf contains only a single observation.

More splits $\Rightarrow$ better training fit, not necessarily better test fit.

Two ways to control complexity:

- **Pre-pruning:** restrict growth.
- **Post-pruning:** grow the tree, then remove branches.

Tree Instability

Machine 12 has low vibration and is labeled failure.
Suppose that label was incorrect and changes from 1 to 0.
The root split at vibration 2.7 now gives

$$L : (7 \text{ normal}, 0 \text{ failure}),$$

$$R : (0 \text{ normal}, 5 \text{ failure}).$$

Both children are pure.
The temperature branch is no longer needed.

Main idea

Changing one training observation can change the structure of a decision tree.

Depth Selection

Illustrative results for three maximum depths:

Maximum depth	1	2	5
Training error	0.08	0.00	0.00
Validation error	0.16	**0.13**	0.22

A deeper tree may achieve the same zero training error but worse validation performance.

Main idea

Depth controls tree flexibility, much as polynomial degree controls curve flexibility.

Pre-Pruning

Set minimum leaf size to 2.
The proposed second split creates leaves of sizes

$$6 \quad \text{and} \quad 1.$$

Since

$$\boxed{1 < 2},$$

the split is rejected.
The low-vibration node remains one leaf and makes one training error.
A simpler model may generalize better.

Post-Pruning

Instead of stopping growth early, first construct a relatively large tree.
Then compare:

- keeping a subtree with several leaves;
- replacing the subtree with one leaf.

Removing a branch generally increases training error but reduces complexity.
We need a criterion to balance both effects.

Cost-Complexity Pruning

For a tree T, define

$$\boxed{R_\alpha(T) = R(T) + \alpha|T|.}$$

Here:

- $R(T)$: training loss;
- $|T|$: number of leaves;
- $\alpha \geq 0$: penalty per leaf.

For our following example, $R(T)$ is the **number of training misclassifications**.
This differs from Gini, which we used to select splits.

Candidate Subtrees

Compare three trees from the machine example.

Tree	Errors	Leaves	Objective
Full tree	0	3	3α
First split only	1	2	$1 + 2\alpha$
Root only	6	1	$6 + \alpha$

The root-only tree makes six errors because both classes have six observations.
The first-split tree makes one error in its low-vibration leaf.

Pruning Penalty

For $\alpha = 0.5$:

$$R_{0.5}(T_3) = 1.5,$$
$$R_{0.5}(T_2) = 2,$$
$$R_{0.5}(T_1) = 6.5.$$

Choose the three-leaf tree.
For $\alpha = 2$:

$$R_2(T_3) = 6, \quad R_2(T_2) = 5, \quad R_2(T_1) = 8.$$

Choose the two-leaf tree.

Larger α can favor a smaller tree.

Pruning Thresholds

The three-leaf and two-leaf objectives tie when

$$3\alpha = 1 + 2\alpha \quad \Longrightarrow \quad \boxed{\alpha = 1}.$$

The two-leaf and one-leaf objectives tie when

$$1 + 2\alpha = 6 + \alpha \quad \Longrightarrow \quad \boxed{\alpha = 5}.$$

α	$0 \le \alpha < 1$	$1 < \alpha < 5$	$\alpha > 5$
Preferred leaves	3	2	1

At $\alpha = 1$ and $\alpha = 5$, adjacent candidates tie.

Weakest-Link Pruning

For a subtree T_t rooted at node t, define

$$\boxed{\alpha_t = \frac{R(t) - R(T_t)}{|T_t| - 1}.}$$

Here:

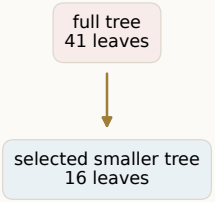
- $R(t)$: loss after replacing the subtree with one leaf;
- $R(T_t)$: current subtree loss;
- $|T_t|$: number of leaves in the subtree.

For the low-vibration branch:

$$\boxed{\alpha_t = \frac{1 - 0}{2 - 1} = 1.}$$

This is the first branch pruned as the penalty increases.

Pruning Illustration



Full tree
more rules

Pruning
replace a subtree

Smaller tree
fewer leaves

Takeaway. Pruning can improve stability by removing branches that add little predictive value.

Validation and interpretation

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Cross-Validation

The tree that best fits training data may not be best for new machines.
A single validation split can also be unstable with limited data.
Cross-validation repeatedly:

1. holds out one part of the training data;
2. fits the tree on the remaining observations;
3. evaluates on the held-out part.

Select complexity using held-out performance.

Keep the final test set untouched.

Tree-Size Comparison

Suppose validation gives the following illustrative errors:

Candidate	Leaves	CV error
Full tree	3	0.19
Pruned tree	2	**0.14**
Root only	1	0.28

The lowest CV error is 0.14.

Select the two-leaf tree.

Its training error may be higher, but its estimated new-data error is lower.

CV Calculation

Suppose four equal-size validation folds give these error rates:

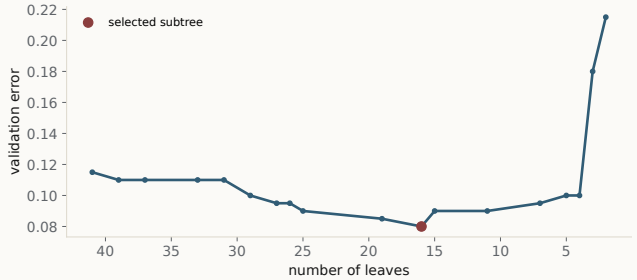
Tree	F1	F2	F3	F4	Mean
Full	.15	.22	.18	.21	.19
2 leaves	.10	.18	.12	.16	**.14**
Root only	.25	.30	.27	.30	.28

For the two-leaf tree:

$$\boxed{CV = \frac{0.10 + 0.18 + 0.12 + 0.16}{4} = 0.14.}$$

All candidates use the same folds for a fair comparison.

Validation Curve



Horizontal axis
tree size

Vertical axis
validation error

Preferred point
lowest error

Takeaway. Validation may favor an intermediate tree rather than the largest or smallest.

Train, Validate, Test

1. **Training:** estimate candidate tree splits.
2. **Validation / CV:** choose depth, leaf size, and pruning penalty.
3. **Final test:** evaluate the selected model once.

Main idea

Repeatedly tuning a model using the final test results turns the test set into another validation set.

Class Imbalance

Suppose only 5 of 100 machines truly fail.
Always predicting normal gives 95% accuracy, but detects no failures.
A different tree produces

$$TP = 4, \quad FN = 1, \quad FP = 2, \quad TN = 93.$$

Then

$$\boxed{\text{Accuracy} = 97\%},$$

$$\boxed{\text{Recall} = \frac{4}{5} = 0.80}, \qquad \boxed{\text{Precision} = \frac{4}{6} \approx 0.667}.$$

Choose metrics that reflect the application.

Trees and Logistic Regression

	Logistic regression	**CART**
Model	linear score, sigmoid	if–then rules
Boundary	linear in original features	axis-aligned pieces
Interactions	need feature construction	arise through branching
Complexity	regularization	stopping and pruning

Main idea

Choose methods using validation, interpretation, and the cost of errors.

Classification and Regression

	Classification tree	**Regression tree**
Response	class label	number
Leaf output	class / proportion	mean
Split criterion	Gini / entropy	squared error
Complexity	depth, leaves, pruning	depth, leaves, pruning

Both use recursive partitioning but different leaf predictions and split criteria.

Leaf Calibration

Suppose a tree leaf predicts

$$\widehat{p}_1 = 0.8$$

from its training class counts.
On a separate evaluation set, 20 machines reach this leaf, and 12 actually fail:

$$\boxed{\text{Observed failure frequency} = \frac{12}{20} = 0.60.}$$

The reported 0.80 appears overconfident in this small group.

Main idea

A split can improve class separation without producing well-calibrated probabilities.

Pruning Practice

Consider three trees with training misclassification counts:

	Training errors	**Leaves**
T_A	0	4
T_B	2	2
T_C	5	1

Use

$$R_\alpha(T) = R(T) + \alpha|T|, \qquad \alpha = 1.5.$$

Compute the objectives and select the preferred tree.

Practice Solution

The objectives are

$$R_{1.5}(T_A) = 0 + 1.5(4) = 6,$$

$$R_{1.5}(T_B) = 2 + 1.5(2) = \boxed{5},$$

$$R_{1.5}(T_C) = 5 + 1.5(1) = 6.5.$$

Thus,

$$\boxed{\text{Select } T_B}.$$

Classification CART	reduce weighted class impurity
Regression CART	reduce squared error
Pruning	balance loss and tree size
Validation	select complexity on held-out data

Week 10: CART from split calculations to validated trees

Questions and discussion